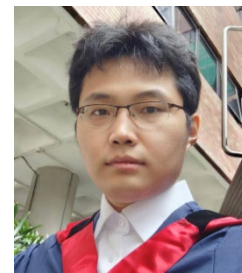


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研究概述

研究方向涵盖三维重建与感知，强化学习，视频生成和具身智能。其中一作/通讯共发表CCF A会议和期刊论文7篇（含1篇TPAMI、3篇CVPR、1篇ICLR、1篇ICML和1篇AAAI）。目前主要关注VLA/WAM和运动生成。

教育经历

香港理工大学	2021.09 – 2026.05
• 计算机系博士	导师: Prof. Bo Yang
西安交通大学	2018.09 – 2021.06
• 数学与统计学硕士	导师: Prof. Huibin Li
电子科技大学	2014.09 – 2018.06
• 数学科学学院学士	

实习经历

华为香港研究院	2025.06 – 2025.08
研究探索3D场景生成和重建	

Selected Papers & Projects

■ VLA/WAM:

概述: 针对VLA/WAM机器人数据匮乏的情况，研究利用自然物体运动数据帮助实现data scaling，降低对机器人数据依赖；同时借助Tracking方法提取3D物体轨迹，并实现多模态生成物体轨迹

1. MotionSlots: Object-Centric 3D Trajectory Prediction as a Scalable Interface for Robot Manipulation

Zihui Zhang, Jinxi Li, Peng Yun, Bo Yang

Ongoing Project

■ 3D Scene Reconstruction:

概述: VGGT等重建方法对Ray的分布敏感泛化性不足，该项目使用RayDF的表达方式提高重建鲁棒性

1. FoundSurface: Feedforward Ray-Based 3D Scene Surface Reconstruction from Unposed Images

Shenxing Wei, Jinxi Li, Zihui Zhang, Ajay Kumar, Wei Wang, Bo Yang

Conference on Neural Information Processing Systems (NIPS), 2026.

■ 3D Scene Understanding:

概述: 3D场景分割大多依赖人工标注，耗时且昂贵，该系列研究基于语义和物体Foundation Models，包括DINO，TRELLIS等，以及PPO强化学习，flow matching生成等技术，实现无监督物体和语义分割

1. FoundObj: Self-supervised Foundation Models as Rewards for Label-free 3D Object Segmentation

Zihui Zhang, Zhixuan Sun, Yafei Yang, Jinxi Li, Jiahao Chen, Bo Yang.

The International Conference on Machine Learning (ICML), 2026

2. GrowSP++: Growing Superpoints and Primitives for Unsupervised 3D Semantic Segmentation

Zihui Zhang, Weisheng Dai, Bo Yang, Bing Wang, Bo Li

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026

3. EvObj: Learning Evolving Object-centric Representations for 3D Instance Segmentation without Scene Supervision

Jiahao Chen, Zihui Zhang[†], Yafei Yang, Jinxi Li, Shenxing Wei, Zhixuan Sun, Bo Yang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [†] indicates corresponding author

4. GrabS: Generative Embodied Agent for 3D Object Segmentation without Scene Supervision

Zihui Zhang, Yafei Yang, Hongtao Wen, Bo Yang
International Conference on Learning Representations (ICLR), 2025 (Spotlight Top 5%)

5. LogoSP: Local-global Grouping of Superpoints for Unsupervised Semantic Segmentation of 3D Point Clouds

Zihui Zhang, Weisheng Dai, Hongtao Wen, Bo Yang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

6. GrowSP: Unsupervised Semantic Segmentation of 3D Point Clouds

Zihui Zhang, Bo Yang, Bing Wang, Bo Li
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023

■ **Video Generation:**

概述: 针对视频新视角生成任务，将其建模为初始噪声挑选和修正过程，无需昂贵后训练

1. NeoMap: Training-free Novel-View Synthesis from Single Images and Videos

Jinxi Li, Tianyi Zhang, Yafei Yang, Zihui Zhang, Peng Huang, Koon Wing Macgyver Lin, Bo Yang
European Conference on Computer Vision (ECCV), 2026

■ **Visual Physical Learning:**

概述: 面向Physical AI的发展需求，利用Unreal Engine制作符合物理规则的大规模benchmark，拥有150K 3D场景和200万多模态视频，涵盖70+类物理现象，覆盖力学，光学，流体和磁力等物理现象。

1. PhysInOne: Visual Physics Learning and Reasoning in One Suite

Siyuan Zhou, Hejun Wang, Hu Cheng, Jinxi Li, Dongsheng Wang, Junwei Jiang, Yixiao Jin, Jiayue Huang, Shiwei Mao, Shangjia Liu, Yafei Yang, Hongkang Song, Shenxing Wei, Zihui Zhang, Bing Wang, Zhihua Wang, Chuhang Zou, Bo Yang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

2. Voken: Feed-Forward Velocity Token as Motion Modality

Jinxi Li, Zihui Zhang, Bo Yang
Ongoing Project

Service

Reviewer for IJCV, TNNLS, TCSVT, CVPR, ICCV, ECCV, NeurIPS, ICML

Awards

- [2026] 优秀博士毕业论文
- [2025] GSCS&ICUI 2025 最佳演讲.
- [2021] 西安交通大学优秀毕业生，毕业论文.
- [2018] 电子科技大学优秀毕业生，毕业论文.